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CHAPTER 14 — FULL TRANSLATION PIPELINE WALKTHROUGH (END-TO-END)

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A maximum-detail, RTOS/QNX-style, end-to-end walkthrough of the complete ARMv8 translation pipeline, from VA generation to final PA access, including Stage-1 + Stage-2 integration, TLB behavior, descriptor interpretation, attribute merging, and execution permission enforcement.

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14.1 — Purpose of This Chapter

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This chapter provides a **single, unified, end-to-end walkthrough** of the entire ARMv8 MMU translation pipeline.

It integrates:

- VA generation
- TTBR selection
- Stage-1 TLB lookup
- Stage-1 page table walk
- IPA generation
- Stage-2 TLB lookup
- Stage-2 page table walk
- PA generation
- permission merging
- attribute merging
- shareability merging
- TLB insertion
- final memory access

This is the “big picture” chapter that ties together everything from Chapters 1–13.

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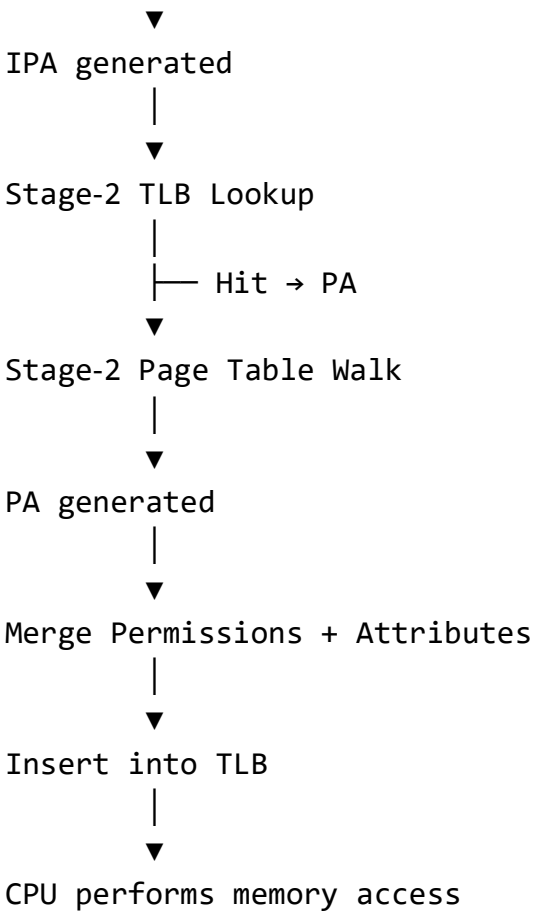
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14.2 — High-Level Pipeline Diagram

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```
CPU generates VA
        │
        ▼
Stage-1 TLB Lookup
        │
        ├── Hit → IPA
        ▼
Stage-1 Page Table Walk
        │
```



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14.3 — Step 1: CPU Generates Virtual Address (VA)

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VA is generated by:

- instruction fetch
- load/store
- atomic operations
- stack operations
- PC-relative addressing
- pointer arithmetic

VA must be:

- canonical
- aligned
- within TTBR0/TTBR1 region

If VA is non-canonical → **Address Size Fault**.

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14.4 — Step 2: TTBR Selection

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MMU selects TTBR based on VA:

```
if VA in TTBR0 region:
    use TTBR0_EL1
else:
    use TTBR1_EL1
```

TTBR0_EL1:

- user space

- ASID-tagged
- per-process

TTBR1_EL1:

- kernel space
- global
- shared across processes

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14.5 — Step 3: Stage-1 TLB Lookup

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TLB lookup requires:

```
VA + ASID + VMID
```

If hit:

- permissions checked
- attributes applied
- IPA returned

If miss:

- Stage-1 page table walk begins.

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14.6 — Step 4: Stage-1 Page Table Walk

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Walk steps:

1. Read L1 descriptor
2. If table → read L2
3. If table → read L3
4. If page/block → final mapping
5. Extract descriptor fields
6. Produce IPA

Walk cache accelerates steps 1–3.

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14.7 — Step 5: Stage-1 Descriptor Interpretation

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Descriptor defines:

- IPA_base
- AP (permissions)
- PXN/UXN (execute permissions)
- SH (shareability)
- AttrIndx → MAIR_EL1 (memory attributes)
- AF (access flag)

If AF = 0 → **Access Flag Fault**. If AP denies access → **Permission Fault**. If PXN/UXN denies execution → **Execute Fault**.

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14.8 — Step 6: IPA Generation

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IPA is computed as:

```
IPA = IPA_base + VA[11:0]
```

IPA is not the real physical address.

IPA is the input to Stage-2.

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14.9 — Step 7: Stage-2 TLB Lookup

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TLB lookup requires:

```
IPA + VMID
```

If hit:

- permissions checked
- attributes applied
- PA returned

If miss:

- Stage-2 page table walk begins.

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14.10 — Step 8: Stage-2 Page Table Walk

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Walk steps:

1. Read L1 descriptor
2. If table → read L2
3. If table → read L3
4. If page/block → final mapping
5. Extract descriptor fields
6. Produce PA

Walk cache accelerates steps 1–3.

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14.11 — Step 9: Stage-2 Descriptor Interpretation

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Descriptor defines:

- PA_base
- S2AP (permissions)
- S2PXN/S2UXN (execute permissions)
- S2SH (shareability)
- S2AttrIndx → MAIR_EL2 (memory attributes)
- AF (access flag)

If S2AP denies access → **Stage-2 Permission Fault**. If S2PXN/S2UXN denies execution → **Stage-2 Execute Fault**.

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14.12 — Step 10: PA Generation

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PA is computed as:

```
PA = PA_base + IPA[11:0]
```

PA is the real physical address used by hardware.

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14.13 — Step 11: Permission Merging

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Final permissions = intersection of Stage-1 and Stage-2.

Example

- Stage-1 AP = read/write

- Stage-2 S2AP = read-only

Final = read-only

Hypervisor always wins.

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14.14 — Step 12: Attribute Merging

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Final attributes = Stage-2 overrides Stage-1.

Example

- Stage-1 AttrIndx = Normal WB
- Stage-2 AttrIndx = Device-nGnRnE

Final = Device-nGnRnE

Hypervisor always wins.

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14.15 — Step 13: Shareability Merging

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Final shareability = Stage-2 overrides Stage-1.

Example

- Stage-1 SH = Non-shareable
- Stage-2 SH = Inner Shareable

Final = Inner Shareable

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14.16 — Step 14: TLB Insertion

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TLB entry contains:

- VA → IPA → PA
- merged permissions

- merged attributes
- ASID
- VMID
- global/non-global flag

TLB entry is inserted into:

- ITLB (instruction)
- DTLB (data)
- UTLB (unified)

Depending on access type.

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14.17 — Step 15: CPU Performs Memory Access

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CPU uses:

- final PA
- final permissions
- final attributes
- final shareability

Memory access proceeds:

- load/store
- instruction fetch

- atomic operation
- device access

If any rule is violated → fault.

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14.18 — Full End-to-End Example

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Given:

```
VA = 0x0000_7FFF_1234_5678
```

Stage-1

- TTBR0_EL1 selected
- L1 → table
- L2 → block
- IPA_base = 0x0000_2000_0000_0000
- AP = read/write
- SH = Inner Shareable
- AttrIndx = Normal WB

IPA = IPA_base + offset

Stage-2

- TTBR0_EL2 selected

- L1 → table
- L2 → block
- PA_base = 0x4000_0000_0000_0000
- S2AP = read-only
- S2SH = Inner Shareable
- S2AttrIndx = Device-nGnRnE

PA = PA_base + offset

Final Permissions

read-only

Final Attributes

Device-nGnRnE

Final Shareability

Inner Shareable

Final Result

CPU performs a read-only access to device memory.

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14.19 — End-to-End Fault Example

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Given:

- Stage-1 AP = read/write
- Stage-2 S2AP = no access

Result:

- Stage-2 Permission Fault
- ESR_EL2 DFSC = 0x23
- FAR_EL2 = IPA
- HPFAR_EL2 = IPA_base

Hypervisor handles the fault.

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14.20 — Chapter Summary

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This chapter covered:

- full end-to-end translation pipeline
- TTBR selection
- Stage-1 TLB lookup
- Stage-1 walk
- IPA generation
- Stage-2 TLB lookup
- Stage-2 walk
- PA generation
- permission merging
- attribute merging
- shareability merging

- TLB insertion
- final memory access
- full examples
- fault examples